

MICHAEL RICHTER

804-893-0534 • michaelrichterswe@gmail.com • linkedin.com/in/michaelrichterswe • michaelrichterswe.com

EXPERIENCE

Capital One, Richmond, VA

AUGUST 2020 – Present

Senior Associate Software Engineer

- Owned and delivered 20+ production AWS Lambda-based services supporting data transformation, secure file delivery to external partners (FIS), and REST APIs for downstream financial systems
- Architected and implemented a message-queue-driven system (AWS SQS) to prevent duplicate processing and ensure idempotent workflows across distributed services
- Built and scaled a serverless reconciliation pipeline processing ~1,000 financial wire transactions per day between internal systems and external vendor platforms (FIS) using AWS Lambda and DynamoDB (NoSQL schema and GSI design)
- Led the end-to-end delivery of a serverless compliance component that encrypted external documents using PGP, eliminating duplicate uploads and saving 30+ associate hours per month
- Standardized and automated AWS infrastructure configuration (S3 policies, triggers, cross-region deployments) across 15 production buckets, reducing setup errors and maintenance overhead
- Designed and automated a Python (Pandas) data ingestion pipeline integrated with internal platform APIs, improving accuracy and efficiency by ~90% of a previously manual process
- Optimized Java-based APIs with pagination and performance enhancements to support large-scale loan data retrieval while maintaining database performance
- Validated system scalability through load and stress testing (Apache JMeter), identifying database bottlenecks prior to production releases
- Validated cross-region resiliency through annual disaster recovery exercises across AWS services
- Mentored and coached junior and mid-level engineers through onboarding, code reviews, and design discussions, improving team delivery speed and code quality

OpenDSA, Blacksburg, VA

August 2019 – December 2019

Undergraduate Research Assistant

- Developed data structure visualizations for an open-source textbook programmed in JavaScript
- Collaborated with a team of 7 to create a comprehensive textbook on programming language constructs, created for 200 students to use throughout each semester

Old Dominion Freight Line, Thomasville, NC

May 2019 – August 2019

Software Engineer Intern

- Developed REST microservices using Spring Boot in a production environment, supporting freight delivery workflows
- Implemented integration tests and mock services, creating 50+ test cases to validate service behavior prior to deployment

EDUCATION

Georgia Tech, Atlanta, GA

EXPECTED GRADUATION: DECEMBER 2028

Master of Science in Computer Science, Specialization in Artificial Intelligence

Virginia Tech, Blacksburg, VA

GRADUATED: MAY 2020

Bachelor of Science in Computer Science, Minor in Mathematics

SKILLS

Languages: Python, Java, JavaScript, SQL

Cloud / Infra: AWS (Lambda, EC2, S3, RDS, SQS, SNS, IAM, DynamoDB)

Tools: Claude Code, Windsurf, GitHub Copilot, Git, Jenkins (CI/CD), Docker, Unix/Linux, JIRA

Certifications: AWS Certified Cloud Practitioner, Anthropic Claude Code, Anthropic MCP